

CIS 6200 Course Project Guidelines

Fall 2026 · Instructor: Surbhi Goel · TA: Berkan Ottlik · <https://www.surbhigoel.com/cis6200>

1 The project

Your project is original research related to the course topics. Your project can be theoretical, empirical, or a mix. You can prove a new result building on a paper we read, define a new model and study it, simplify an existing result in a non-trivial way, design experiments that test a theoretical prediction, and more.

Your groups can have up to three people, and expectations scale with group size. Your proposal must be approved by Surbhi and you will get feedback after submitting. We also encourage scheduling office hours with us. You may build on your current research if it's clearly connected to the course.

2 Deliverables and dates

Your project counts for 65% of your grade.

Deliverable	% of grade	Due	What we expect
Team and Project Title	0%	Oct 1	Fill in this form with your team information, a working title and a TLDR for your project.
Proposal	5%	Oct 15	Write at most two pages. References can go over the limit. State your research question and explain why it matters. Summarize the most relevant prior work and say what it leaves open. Describe your plan with the first concrete step, what you expect it to show, and what you will try if it fails. If you are in a group, say how you will divide responsibilities.
Mid-semester Talk	15%	Nov 5	Present your progress for a few minutes (exact time will depend on the number of projects, it will be announced later). Slides or the whiteboard are both fine. State your question, describe what you have tried, and show your preliminary results, including the negative ones. End with your plan for the remaining weeks.
Final Talk	25%	Dec 1–3	Present your entire work (exact time will depend on the number of projects and be about twice as long as the mid-semester talk, it will be announced later). Slides or the whiteboard are both fine. Present the problem, your approach, your main results, and your conclusions, and say what you would do next.
Final Report	20%	Dec 14	Write at most six pages, plus references and appendices, using the NeurIPS style files . Open with an abstract and introduction that states your question and your contribution. Then cover related work, your setup, your results, and a discussion of limitations. Put full proofs and additional experiments in appendices, and include a statement of how you used AI tools. You do not need to fill all six pages.

3 What we are looking for

Be ambitious. This course is a good place to try something that might not work. If your ambitious project does not fully succeed, that is fine, just explain what you tried, what you learned, and what remains open. We want you to ask a real question, attack it seriously, and report honestly what you found.

Size. To calibrate the size of your project, look at papers from workshops such as [HiLD](#) (ICML), [MOSS](#) (ICML), [M3L](#) (NeurIPS), and [ME-FoMo](#) (ICLR). Each develops one self-contained idea in a few pages.

Compute. Assume that we cannot provide compute.¹ If your project is empirical, plan to run it on your laptop, on Colab, or on resources you already have.

A literature review is not a project. Your report must contain something you produced: a theorem, a model, a simpler proof, an experiment, a counterexample.

4 Getting started

Finding a topic. Start with the reading list. Pick a paper you found interesting and pull on it. Look it up on Google Scholar to see who has cited it, read what the same authors have done since, and check its future work section, where authors are usually candid about what they could not make work. Ask what an assumption is doing, whether a result survives without it, or whether a prediction it makes holds up when you implement the idea. The accepted papers at the workshops linked above, are also worth browsing for recent work in the field. You can also read broader surveys of LLM theory such as [Simon et al. \(2026\)](#) and [Biderman et al. \(2026\)](#) to find open questions we do not cover in class. You can also work backwards from a real problem you have, or one you think AI might cause or solve.

On doing research. Here are some resources that might help you do research: Hamming’s *You and Your Research* ([Hamming, 1986](#)), Barak’s *Advice for the Budding Theorist* ([Barak, 2015](#)), Alon’s *How to Choose a Good Scientific Problem* ([Alon, 2009](#)), and Olah’s *Research Taste Exercises* ([Olah, 2021](#)).

AI tools. You can use AI tools for every part of the project, e.g., brainstorming, math, code, writing. However, you are responsible for everything you submit. You must be able to explain any part of it when we ask, and you must include in each deliverable a short statement of how you used these tools.

Talk to us. We suggest your teams to meet with one of us before your proposal is due and again while working on the project. We love talking to you, helping you with your projects, and answering any questions about the class content or other stuff! You can sign up for office hours through the links on Canvas.

References

- Uri Alon. How to choose a good scientific problem. *Molecular Cell*, 35(6):726–728, 2009. URL <https://doi.org/10.1016/j.molcel.2009.09.013>.
- Boaz Barak. Advice for the budding theorist. Windows On Theory (blog), November 2015. URL <https://windowsontheory.org/2015/11/03/advice-for-the-budding-theorist/>.

¹Surbhi is working on acquiring compute for the class but cannot promise anything yet.

Stella Biderman, Mohammad Aflah Khan, Niloofar Miresghallah, Catherine Arnett, Fazl Barez, and Naomi Saphra. Position: Don't just "fix it in post": A science of AI must study training dynamics. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. URL <https://arxiv.org/pdf/2606.06533>.

Richard W. Hamming. You and your research. Transcript of the Bell Communications Research Colloquium Seminar, March 1986. URL <https://gwern.net/doc/science/1986-hamming>.

Chris Olah. Research taste exercises. colah's blog, January 2021. URL <https://colah.github.io/notes/taste/>.

Jamie Simon, Daniel Kunin, Alexander Atanasov, Enric Boix-Adserà, Blake Bordelon, Jeremy Cohen, Nikhil Ghosh, Florentin Guth, Arthur Jacot, Mason Kamb, Dhruva Karkada, Eric J. Michaud, Berkan Ottlik, and Joseph Turnbull. There will be a scientific theory of deep learning. arXiv preprint arXiv:2604.21691, 2026. URL <https://arxiv.org/pdf/2604.21691>.